

Medical Data Science Workflow for Cardiovascular Risk Prediction Using Clinical Patient Data

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1. Executive Summary

Cardiovascular disease is a major global health challenge, and early identification of patients at increased risk can support preventive care and clinical decision-making. This project applies a complete medical data science workflow to develop predictive machine learning models for heart disease classification using patient clinical information.

The study analysed a structured healthcare dataset containing demographic, physiological, and diagnostic variables. The workflow included:

- Data quality assessment
- Exploratory data analysis
- Statistical analysis
- Feature preprocessing
- Machine learning model development
- Model validation
- Hyperparameter optimisation
- Feature importance analysis

Two supervised machine learning approaches were evaluated:

1. Logistic Regression
 - Selected due to interpretability and clinical relevance.
2. Random Forest Classifier
 - Selected due to its ability to capture non-linear relationships and complex interactions between clinical variables.

Model performance was evaluated using accuracy, precision, recall, F1-score, ROC-AUC, confusion matrices, and five-fold cross-validation.

The results demonstrated that both models were able to identify patterns associated with heart disease. Logistic Regression achieved stronger performance on the independent test

set, while Random Forest demonstrated higher cross-validation ROC-AUC, indicating strong generalisation capability.

2. Clinical Objective

The objective of this project was to develop machine learning models capable of predicting the presence of heart disease based on patient clinical characteristics.

The target variable was:

- 1 = Heart disease present
- 0 = Heart disease absent

The goal was to investigate whether machine learning could assist in cardiovascular risk assessment by identifying important clinical predictors.

3. Dataset Description

The dataset contained:

Dataset Characteristic	Value
Number of records	1025
Number of variables	14
Target variable	Heart disease status
Missing values	0
Duplicate records	723

** The dataset contains 14 variables (13 input features and 1 target variable)

```

[1025 rows x 14 columns]
(1025, 14)
<class 'pandas.DataFrame'>
RangeIndex: 1025 entries, 0 to 1024
Data columns (total 14 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   age         1025 non-null   int64
 1   sex         1025 non-null   int64
 2   cp          1025 non-null   int64
 3   trestbps    1025 non-null   int64
 4   chol        1025 non-null   int64
 5   fbs         1025 non-null   int64
 6   restecg     1025 non-null   int64
 7   thalach     1025 non-null   int64
 8   exang       1025 non-null   int64
 9   oldpeak     1025 non-null   float64
10   slope       1025 non-null   int64
11   ca          1025 non-null   int64
12   thal        1025 non-null   int64
13   target      1025 non-null   int64
dtypes: float64(1), int64(13)
memory usage: 112.2 KB

```

After duplicate removal:

Item	Result
Final dataset size	302 patient records

The dataset contained the following clinical features:

Numerical Variables

- Age
- Resting blood pressure
- Cholesterol
- Maximum heart rate achieved
- ST depression (oldpeak)

Categorical Variables

- Sex
- Chest pain type
- Fasting blood sugar
- Resting ECG
- Exercise-induced angina
- Slope
- Number of major vessels
- Thalassemia

Data Source: Dataset source: Publicly available cardiovascular disease dataset from Kaggle website. The dataset contains anonymized patient-level clinical variables and does not include personally identifiable information.

4. Data Quality Assessment

Missing Data

A complete missing value assessment was performed.

Result:

- No missing values were identified.

Therefore, no imputation strategy was required.

Duplicate Detection

Duplicate records were identified and removed.

This step was important because repeated observations can artificially increase model performance and reduce reliability.

After cleaning:

- Dataset contained unique patient observations.
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5. Exploratory Data Analysis

Exploratory analysis was performed to understand patient characteristics and identify relationships between clinical variables.

The analysis included:

- Feature distributions (for example, Chol)
- Target class distribution
- Correlation analysis
- Outlier detection

The target variable distribution was examined to ensure appropriate model evaluation.

A stratified train-test split was applied to maintain similar disease/non-disease proportions between training and testing datasets.

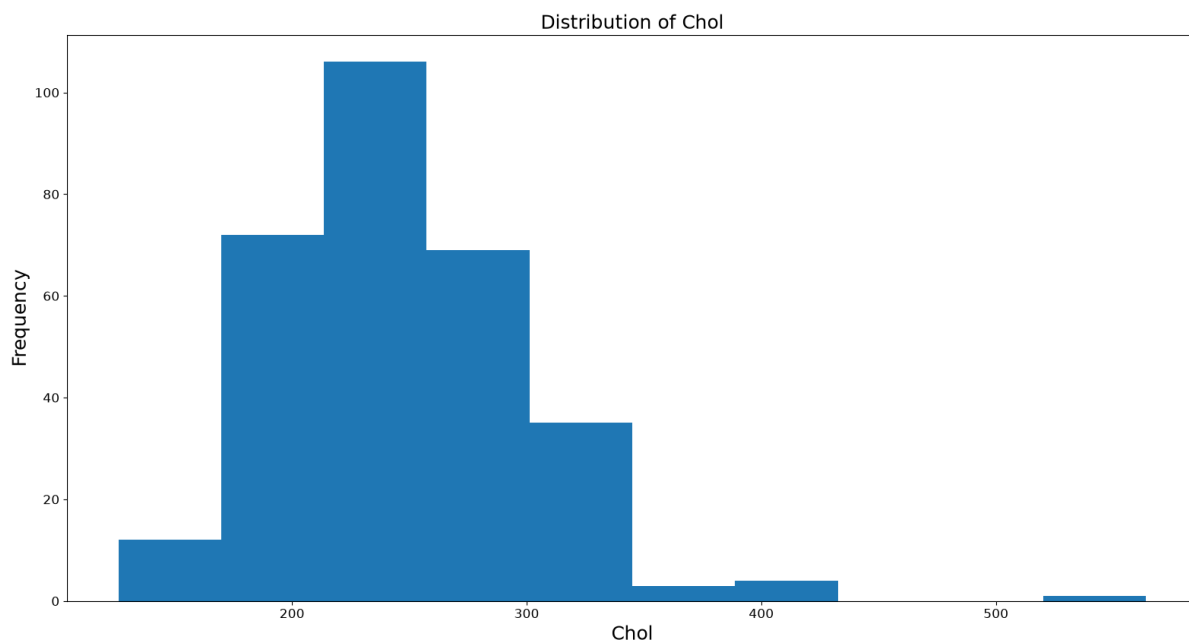


Figure.1-Distribution of Cholesterol (Chol). It is skewed with outliers

Outlier Detection

In many machine learning projects, outliers are removed because they may result from measurement errors, data entry mistakes, or sensor-related problems. However, in healthcare datasets, extreme values may represent genuine physiological conditions rather than errors. For example, a high cholesterol value may indicate:

- A patient with dyslipidaemia
- Increased cardiovascular risk

Removing such values could eliminate clinically meaningful patient information from the dataset. Therefore, in this clinical project, outliers were not removed because they may represent important variations in patient health characteristics.

6. Correlation Analysis

Correlation analysis was conducted for numerical variables.

The purpose was to:

- Identify relationships between clinical measurements
- Detect possible redundancy between variables
- Understand cardiovascular risk patterns

The analysis included:

- Age
- Blood pressure
- Cholesterol
- Maximum heart rate
- ST depression

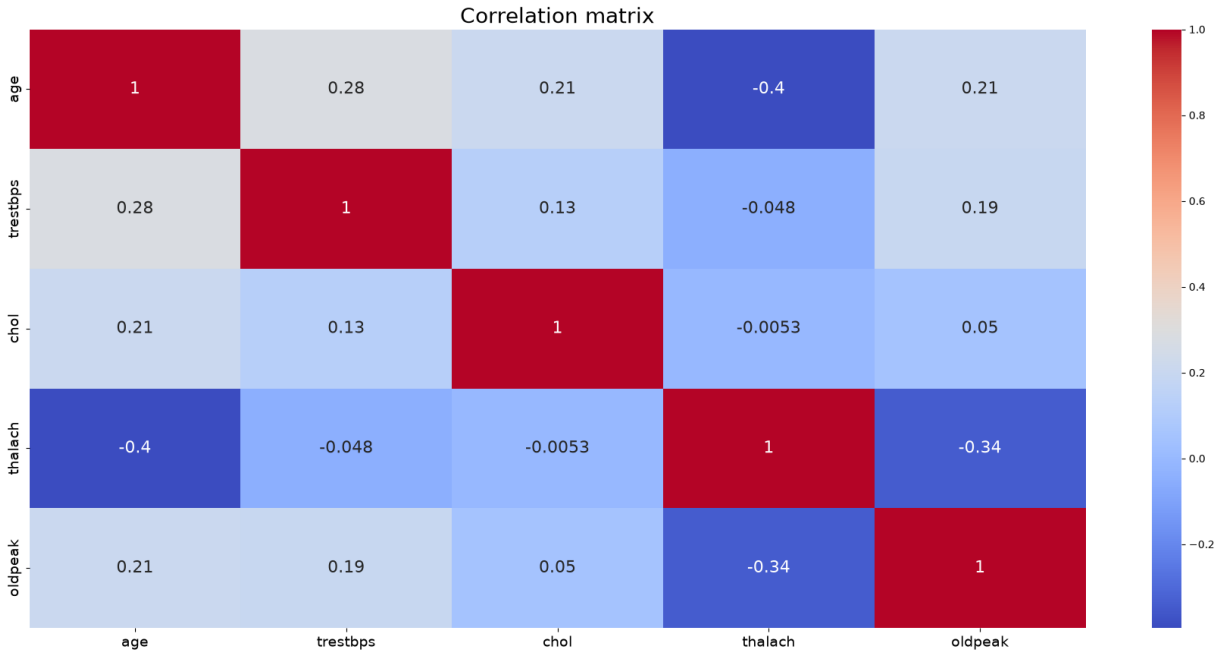


Figure.2-Correlation among the numerical variables

7. Statistical Analysis

Statistical hypothesis testing was performed to evaluate relationships between clinical variables and heart disease outcomes.

Numerical Variables

Independent t-tests were used.

Hypothesis:

- H0: There is no difference between patients with and without heart disease.
- H1: The clinical feature differs significantly between groups.

Significance level: $\alpha = 0.05$

Table 1- P-value of each categorical variable associated with heart disease in the dataset

Numerical Feature	Mean (Disease=1)	Mean (Disease=0)	t-statistic	p-value	Significant?
Age	52.59	56.6	-3.994	8.18×10^{-5}	Yes
Resting BP (trestbps)	129.25	134.4	-2.529	0.012	Yes
Cholesterol (chol)	242.64	251.09	-1.425	0.1553	No
Maximum Heart Rate (thalach)	158.38	139.1	7.904	6.90×10^{-14}	Yes
ST Depression (oldpeak)	0.59	1.59	-7.902	1.38×10^{-13}	Yes

Table 1 presents the results of the t-test, indicating which numerical variables are significantly associated with heart disease and which are not. The significance threshold for this study was set at 0.05. After conducting an independent t-test, the results showed that cholesterol levels were not significantly associated with heart disease in this dataset, with a p-value of 0.1553. In contrast, age was found to be statistically significantly associated with heart disease because the mean age differed significantly between the two groups. Since the p-value was less than 0.05 ($p < 0.05$), the null hypothesis was rejected.

Although age is a well-established cardiovascular risk factor, this dataset does not fully demonstrate the expected population-level pattern. This may be because the dataset consists of selected patients who have already undergone clinical evaluation for cardiac disease rather than representing a general population sample.

Categorical Variables

Chi-square tests were applied to determine whether categorical variables were associated with heart disease.

This approach evaluates whether variables such as chest pain type or exercise-induced angina are statistically related to disease status.

The hypothesis tested was:

➤ H_0 (Null hypothesis):

There is no association between the categorical variable and heart disease status.

➤ H_1 (Alternative hypothesis):

The categorical variable is associated with heart disease status.

Significance level: $\alpha=0.05$

Table 2- P-value of each categorical variable associated with heart disease in the dataset

Categorical Variable	Chi-square Statistic	Degrees of Freedom	p-value	Significant Association?
Sex (sex)	78.863	1	6.66×10^{-19}	Yes
Chest pain type (cp)	280.982	3	1.30×10^{-60}	Yes
Fasting blood sugar (fbs)	1.513	1	0.2186	No
Resting ECG (restecg)	35.784	2	1.70×10^{-8}	Yes
Exercise-induced angina (exang)	194.816	1	2.83×10^{-44}	Yes
Slope (slope)	155.873	2	1.42×10^{-34}	Yes
Thalassemia (thal)	280.331	3	1.80×10^{-60}	Yes

The p-values of the variables presented in Table 2 are less than 0.05, indicating that these variables are statistically significantly associated with heart disease in this dataset. For example, sex is significantly associated with heart disease because the distribution of heart disease cases differs substantially between males and females. In contrast, fasting blood sugar is not significantly associated with heart disease in this dataset, as its p-value is greater than 0.05. This suggests that sex may represent an important cardiovascular risk factor, as biological, hormonal, and lifestyle-related differences between males and females can influence cardiovascular risk.

8. Data Preprocessing

Feature Scaling

Standardisation was applied for Logistic Regression.

The transformation ensured:

- Mean = 0
- Standard deviation = 1

Scaling was applied because Logistic Regression is sensitive to feature magnitude.

Random Forest was trained using the original feature values because tree-based algorithms do not require scaling.

9. Machine Learning Models

9.1 Logistic Regression

Logistic Regression was selected because it provides:

- Interpretability
- Probability-based predictions
- Clinical transparency

This is valuable in healthcare where understanding decision factors is important.

9.2 Random Forest

Random Forest was selected because it:

- Captures non-linear patterns
 - Models interactions between variables
 - Provides feature importance ranking
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10. Model Evaluation Results

Independent Test Set Performance

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	80.3%	80.0%	84.8%	82.4%	0.871
Random Forest	75.4%	76.5%	78.8%	77.6%	0.861

11. Confusion Matrix Analysis

Logistic Regression

The model correctly identified 28 out of 33 disease cases.

The false negative rate was relatively low, which is important in healthcare because missed disease cases may delay treatment.

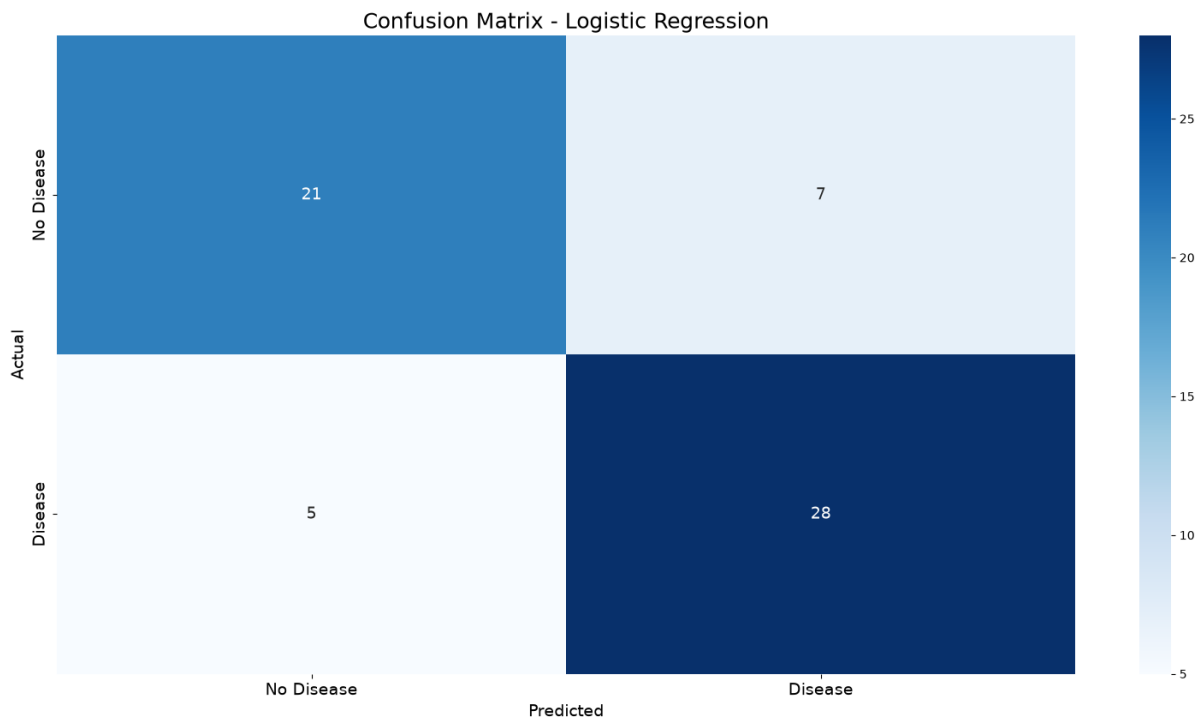


Figure.3-Confusion matrix for Logistic Regression

True Positive (TP)

Patients who actually had heart disease and were correctly identified by Logistic Regression.

These are successful detections.

The model correctly recognised high-risk patients who may require:

- Further cardiac investigation
- Preventive intervention
- Clinical follow-up

A higher TP count contributes to higher sensitivity (recall).

True Negative (TN)

Patients without heart disease who were correctly classified as healthy.

The model successfully avoided unnecessary concern or additional testing for low-risk patients.

A good TN value indicates the model can distinguish healthy patients from diseased patients.

False Positive (FP)

Patients without heart disease who were incorrectly predicted as having disease.

These patients may experience:

- Additional diagnostic testing
- Increased healthcare costs
- Possible anxiety

However, in cardiovascular screening, false positives are generally considered less harmful than false negatives because further investigation can confirm the diagnosis.

False Negative (FN)

Patients who had heart disease but were incorrectly predicted as healthy.

This is the most clinically important error.

A false negative may result in:

- Delayed diagnosis
- Missed treatment opportunity
- Increased cardiovascular risk

Therefore, reducing FN cases is a priority in medical prediction models.

The confusion matrix demonstrated that Logistic Regression was able to correctly classify a substantial proportion of patients with and without heart disease. The model achieved a strong ability to identify disease cases, reflected by its sensitivity/recall performance. However, some false negative predictions remained, indicating that a proportion of patients with underlying disease may not be detected. In a clinical setting, further optimisation may focus on reducing false negatives because missed disease cases carry greater clinical consequences than unnecessary follow-up investigations.

Random Forest

Random Forest also demonstrated good disease detection capability.

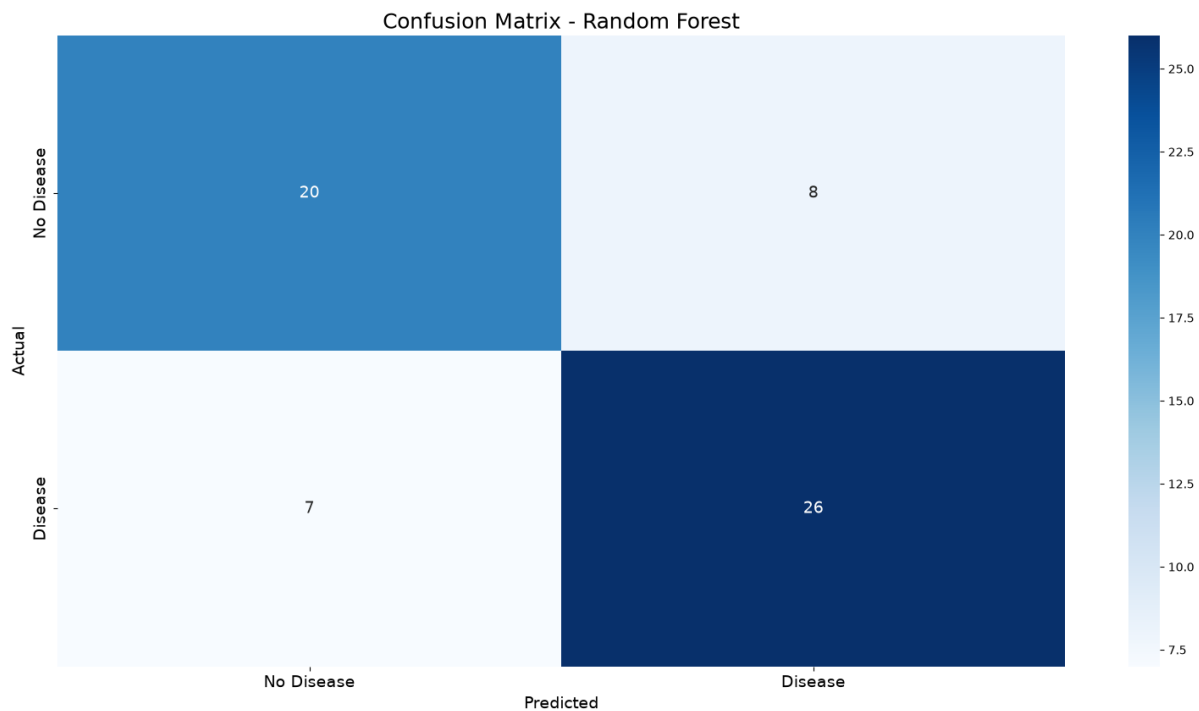


Figure.4-Confusion matrix for Random Forest

True Positive (TP)

Patients with heart disease correctly identified by Random Forest.

The model successfully detected patients whose clinical profiles matched patterns associated with cardiovascular disease. This demonstrates that Random Forest can recognise complex combinations of risk factors.

True Negative (TN)

Patients without heart disease correctly classified. The model successfully identified lower-risk individuals. This reduces unnecessary clinical follow-up.

False Positive (FP)

Healthy patients incorrectly classified as having heart disease.

Although these predictions are incorrect, they may be acceptable in a screening context because they lead to further assessment rather than missed diagnosis.

False Negative (FN)

Patients with heart disease incorrectly classified as healthy. These are the most concerning predictions.

Possible consequences include:

- Failure to identify cardiovascular risk
- Delayed intervention
- Missed opportunity for preventive care

Therefore, recall remains a critical evaluation metric.

The Random Forest confusion matrix showed that the model successfully identified many patients with heart disease while maintaining reasonable classification of non-disease cases. The model's ability to capture complex interactions between clinical variables contributed to its predictive performance. However, false negative cases remain clinically important because missed cardiovascular disease detection may delay appropriate management. Therefore, model evaluation should prioritise sensitivity and risk stratification rather than accuracy alone

12. Cross-Validation Results

Five-fold cross-validation was performed using ROC-AUC.

Model	Mean ROC-AUC
Logistic Regression	0.894
Random Forest	0.911

Random Forest achieved a higher cross-validation score, suggesting stronger predictive stability across different data splits. Furthermore, Random Forest achieved a higher cross-validation ROC-AUC compared with Logistic Regression, indicating superior discriminative performance. These results demonstrate that Random Forest can capture patterns within the dataset more effectively and maintain consistent performance across different subsets of the data. The strong cross-validation performance suggests that the learned

patterns are reproducible and that Random Forest has stronger generalisation ability compared with Logistic Regression.

13. Hyperparameter Optimisation

GridSearchCV was used to identify optimal model configurations.

Logistic Regression

Best parameters:

Parameter	Value
C	0.1
Maximum iterations	2000

Best cross-validation accuracy:

82.6%

Random Forest

Best parameters:

Parameter	Value
Number of trees	200
Maximum depth	3

Best cross-validation accuracy:

85.5%

14. Feature Importance Analysis

Random Forest feature importance identified the strongest contributors to heart disease prediction.

Rank	Feature	Importance
1	Chest pain type (cp)	0.174
2	Maximum heart rate (thalach)	0.132
3	Number of vessels (ca)	0.106
4	ST depression (oldpeak)	0.097
5	Thalassemia	0.090
6	Age	0.083

The results indicate that clinical indicators related to cardiac symptoms, exercise response, and physiological measurements contributed strongly to prediction.

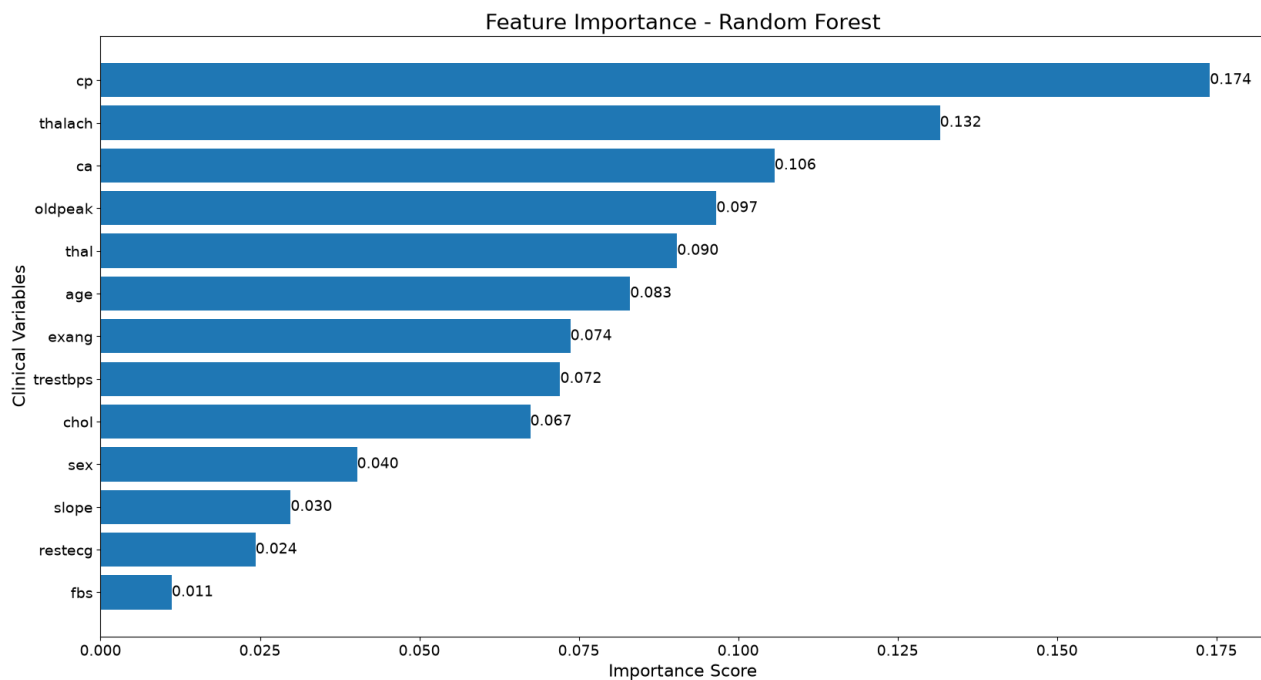


Figure.5-Feature importance captured by Random Forest

It is clear from Fig.5 that Chest pain type (cp) was one of the most influential predictors. The model learned that different chest pain patterns are associated with different probabilities of heart disease.

15. Clinical Interpretation

The developed machine learning models demonstrate the potential of artificial intelligence for cardiovascular risk prediction.

Key findings:

- Logistic Regression provided strong predictive performance with high interpretability.
- Random Forest provided strong generalisation capability through cross-validation.
- Clinical variables related to chest pain, heart rate response, and exercise-related measurements were important predictors.

These models could potentially support clinicians by providing additional risk information during patient assessment.

However, machine learning predictions should be considered decision-support tools rather than replacements for clinical judgement.

16. Limitations

Several limitations should be considered:

1. The dataset size is relatively limited.
 2. The study requires external validation using independent clinical populations.
 3. Dataset characteristics may not represent all demographic groups.
 4. Before clinical deployment, additional evaluation is required, including:
 - Model calibration
 - Bias assessment
 - Prospective validation
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17. Conclusion

This project demonstrates a complete medical data science workflow for heart disease prediction.

The analysis successfully integrated:

- Healthcare data cleaning

- Statistical investigation
- Machine learning modelling
- Model validation
- Explainability analysis

Both Logistic Regression and Random Forest demonstrated the ability to predict heart disease from patient clinical information.

Future work should focus on external clinical validation, explainable AI methods, and integration into clinical decision-support systems.